

2023/2022 Maths Paper

Maths 2

(a) Given n is odd, show that

$$n^2 + 4n + 3$$

is a multiple of 8 for n ∈ N.

answer:

if n is odd, we can write n as (2x + 1), where x ∈ N,

then we substitute (2x + 1) into the system:

$$(2x+1)^2 + 4(2x+1) + 3 =>$$
$$4x^2 + 4x + 1 + 8x + 4 + 3 =>$$
$$4x^2 + 12x + 8 =>$$
$$4(x^2 + 3x + 2)$$

- prove x^2 + 3x + 2 is even, i.e. it can be written in the form of 2k.
- finish later

(b) Use proof by contradiction to show that every real number x satisfies the following inequality:

$$|x + \frac{1}{x}| \geq 2$$

answer:

$$|x + \frac{1}{x}| < 2$$

prove this

$$x^2 + 2 + \frac{1}{x^2} < 4$$
$$x^4 - 2x^2 + 1 < 0$$
$$(x^2 - 1)(x^2 - 1) < 0$$
$$(x^2 - 1)^2 < 0$$

a bracket square, i.e. ()^2 is always ≥ 0, the last implication is a contradiction of this, hence

$$|x + \frac{1}{x}| \geq 2$$

is true for all x ⇒ Real number

(c) Determine whether the following argument is valid or invalid. Give a fully justified reason for your answer by defining suitable propositions and/or predicates, stating the argument you are going to check using these and then determining whether the argument is valid or not.

- It is not raining or Alice has her raincoat.
- Alice does not have her raincoat or she does not get wet.
- It is raining or Alice does not get wet.
- Therefore, Alice does not get wet.

(d) Express gcd(252, 356) as a linear combination of 252 and 356, where gcd(a, b) is the greatest common divisor of a and b.

❌ 365=(252*1)+(113*1)\\
252=(113*2)+(26*1)\\
113=(26*4)+(9*1)\\
26=(9*2)+(8*1)\\
9=(8*1)+(1*1)\\
8=(1*8)\\
1\ is\ gcd(252,365)
\\
1=9-(8*1)\\
1=9-(26-(9*2))\\
1=(9*3)-(1*26)\\
1=((113-(26*4))*3)-(1*26)\\
1=(113*3-(26*12)-(1*26)\\
1=(113*3)-26*13\\
1=(113*3)-(252-(113*2)*13)\\
1=(113*3)-(252*13-113*26)\\
1=(113*29)-(252*13)\\
1=((365-252*1)*29)-(252*13)\\
1=365*29-252*42

🔼 I mis-read the question 😢

$$356 = 252 * 1 + 104 * 1$$
$$252 = 104 * 2 + 44 * 1$$
$$104 = 44 * 2 + 16 * 1$$
$$44 = 16 * 2 + 12 * 1$$
$$16 = 12 * 1 + 4 * 1$$
$$12 = 4 * 3$$
$$4 \text{ is the greatest common divsior of } 252 \text{ and } 356 \text{ i.e. } 4 = \gcd(252, 356)$$
$$4 = 16 - 12 * 1$$
$$4 = 16 - (44 - 16 * 2) * 1 = 16 - 44 * 1 + 16 * 2 = 16 * 3 - 44 * 1$$
$$4 = 16 * 3 - 44 * 1$$
$$4 = (104 - 44 * 2) * 3 - 44 * 1 = (104 * 3 - 44 * 6) - 44 * 1 = 104 * 3 - 44 * 7$$
$$4 = 104 * 3 - 44 * 7$$
$$4 = 104 * 3 - (252 - 104 * 2) * 7 = 104 * 3 - (252 * 7 - 104 * 14) = 104 * 3 - 252 * 7 + 104 * 14 = 104 * 17 -$$
$$4 = 104 * 17 - 252 * 7$$
$$4 = (356 - 252 * 1) * 17 - 252 * 7 = 356 * 17 - 252 * 17 - 252 * 7 = 356 * 17 - 252 * 24$$
$$4 = 356 * 17 - 252 * 24$$

(e) Let

$$g(n) = 5^{2n} - 1, n \in N$$

Without using proof by induction, show that g(n) is a multiple of 8.

answer:

idk how to do this one 😭

Discrete Maths

(a) Using proof by induction, prove:

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + ... + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$$

Hint:

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

Proof by induction:

Base Case: n = 1

$$\frac{1}{1(1+1)} = 1 - \frac{1}{1+1}$$
$$\frac{1}{2} = 1 - \frac{1}{2}$$
$$\frac{1}{2} = \frac{1}{2}$$

Yes.

Inductive Case:

Assume case k is true, i.e.

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + ... + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1}$$

is true.

prove case k + 1 is true:

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + ... + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{k+1+1}$$

From our assumption from case k, we can substituted:

$$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+...+\frac{1}{k(k+1)}=1-\frac{1}{k+1}$$

into the above equation:

$$1-\frac{1}{k+1}+\frac{1}{(k+1)(k+2)}=1-\frac{1}{k+2}$$

simplifying the LHS:

$$\begin{aligned}&\frac{(k+1)(k+2)}{(k+1)(k+2)}-\frac{1(k+2)}{(k+1)(k+2)}+\frac{1}{(k+1)(k+2)}\\&=\frac{(k+1)(k+2)-(k+2)+1}{(k+1)(k+2)}\\&=\frac{k^2+3k+2-k-2+1}{(k+1)(k+2)}\\&=\frac{k^2+2k+1}{(k+1)(k+2)}\\&=\frac{(k+1)(k+1)}{(k+1)(k+2)}\\&=\frac{k+1}{k+2}\end{aligned}$$

simplifying the RHS:

$$\begin{aligned}&1-\frac{1}{k+2}\\&=\frac{k+2}{k+2}-\frac{1}{k+2}\\&=\frac{k+2-1}{k+2}\\&=\frac{k+1}{k+2}\end{aligned}$$

Since LHS and the RHS are the same, by the rule of induction the given statement:

$$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+...+\frac{1}{n(n+1)}=1-\frac{1}{n+1}$$

is true for all n, n is an element of positive integers.

(b) Prove that A \ (B \ C) = (A \ B) U (A N C) for all sets A, B, and C.

(c) Let

$$A=\{\frac{p}{2^n}\in\mathbb{Q}|\ p,n\in\mathbb{N},p\geq 1,n\geq 1,\text{p is odd}\}$$

Prove that A together with the multiplication operation is a semigroup, whereas A with the addition operation is not.

(d) How many distinct isomorphism from the complete bipartite graph K_2,3 to itself are there, Justify your answer.

(a) Let L be a regular language. Prove that L*, the Kleene star applied to L, is a regular language.

(b) Let L be the language over the alphabet A = {0, 1} consisting of every string of even length such that the number of occurrences of 0 in the string is odd. For example, 01 is in L, but 0011 is not in L. Draw a deterministic finite state acceptor that recognises language L. Make sure that you label the start state, finishing state, and all the transitions.

(c) Give the specification of a regular grammar in normal form that generates the language L from part (b). Show that indeed all strings in this language are generated by the grammar you gave and no others.

(d) Since you were able to construct a finite state acceptor that recognises L in part (b), the language L must be regular, so it must fulfil the conclusion of the Pumping Lemma. Consider w = 010101. Decompose w into x, u, and y such that `w = xuy`, |`xu`| ≤ 4, u ≠ ε (null string), and `xu^ny` ∈ (is in) L ∀n ≥ 0. Justify your answer.

Probability

(a) Suppose that an ordinary deck of 52 cards is shuffled and the cards are then turned over one at a time until the first ace appear. If the first ace is the 20th card to appear, what is the conditional probability that the following card is:

(i) ace of spades?

(ii) two of clubs?

(b)

(i)A gambler has a fair coin and a two-headed coin in their pocket. They select one of the coins at random; when they flip it, it shows heads. What is the probability that it is the fair coin?

(ii) Suppose the same coin is flipped a second time, and, again, it shows heads. Now what is the probability that it is the fair coin?

(iii) Suppose the same coin is flipped a third time and it shows tails. Now what is the probability that it is the fair coin?

(c)

There are two machines (M1 and M2), one that requires a single AA battery and another that uses two AAA batteries. There are two AA batteries and four AAA batteries available. Suppose the probability that any particular battery works is p and that batteries work or fail independently of one another. Suppose we have to pick one of the machines. For what values of p should we pick the machine M1?

(d)

Two different professors have submitted final exams for duplication. Let X denote the number of typographical errors on the first professor's exam and Y denote the number of such errors on the second exam. Suppose X has a Poisson distribution with parameter $\mu 1$, and Y has a Poisson distribution with parameter $\mu 2$, and X and Y are independent.

(i) What is the joint probability mass function of X and Y ?

(ii) What is the probability that at most one error is made on both exams combined?

(iii) Obtain a general expression for the probability that the total number of errors in the two exams is m (where m is a nonnegative integer).